

Analysis of Asynchronous Sequential Circuit

Step 1: Problem Statement

Step 2: Primitive Flow Table

Step 3: Flow Table

Step 4: Transition Table

Step 5: Logic Diagram (Combination OR Using S-R latch)

Example: Design a asynchronous sequential circuit in gated latch with 2 inputs A and B and output Q.

Whenever B is 1 input A is transferred to Q, when B is 0 output remains in the same even if A changes its value.

State	Input		Output
	A	B	Q
A	0	1	0
B	0	0	0
C	1	0	0
D	1	1	1
E	1	0	1
f	0	0	1

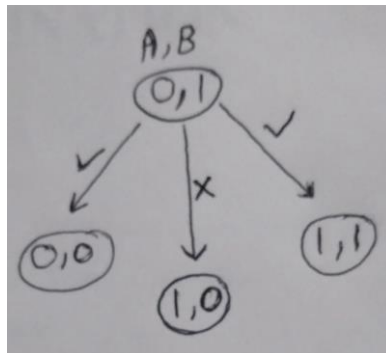
Primitive Flow Table:

(Fundamental Mode Principle– There will be only one bit change, Two bit change not allowed. ie. If inputs A, B = 0, 0 then possible changes are 0, 1 & 1, 0. Another value 1, 1 not allowed, because of two bit changes)

Inputs A B ->	0 0	0 1	1 1	1 0	Remark/Step Follow
State a	b, -	<u>a</u> , 0	d, -	-, -	Inputs A, B = 0, 1 then possible changes are 0, 0 & 1, 1. Another value 1, 0 not allowed, because of two bit changes. These two possible changes are unstable state and same bits are stable state, so mark underline. Write two -, - two changes define.
State b	<u>b</u> , 0	a, -	-, -	c, -	Inputs A, B = 0, 0 then possible changes are 0, 1 & 1, 0. Another value 1, 1 not allowed, because of two bit changes. These two changes are unstable state and same bits are stable state, so mark underline. Write two -, - two changes define.
State C	b, -	-, -	d, -	<u>c</u> , 0	Inputs A, B = 1, 0 then possible changes are 0, 0 & 1, 1. Another value 1, 0 not allowed, because of two bit changes. These two changes are unstable state and same bits are stable state, so mark underline. Write two -, - two changes define.

State d	<u>-</u> , -	a, 0	<u>d</u> , 1	e, -	Inputs A, B = 1, 1 then possible changes are 0, 1 & 1, 0. Another value 0, 0 not allowed, because of two bit changes. These two changes are unstable state and same bits are stable state, so mark underline. Write two -, - two changes define.
State e	f, -	-, -	d, -	<u>e</u> , -	Inputs A, B = 1, 0 then possible changes are 0, 0 & 1, 1. Another value 1, 0 not allowed, because of two bit changes. These two changes are unstable state and same bits are stable state, so mark underline. Write two -, - two changes define.
State f	<u>f</u> , 1	a, -	-, -	e, -	Inputs A, B = 0, 0 then possible changes are 0, 1 & 1, 0. Another value 1, 1 not allowed, because of two bit changes. These two changes are unstable state and same bits are stable state, so mark underline. Write two -, - two changes define.

Example: Inputs A, B = 0, 1 then possible changes are 0, 0 & 1, 1. Another value 1, 0 not allowed, because of two bit changes. These two possible changes are unstable state and same bits are stable state, so mark underline. Write two -, - two changes define.



Inputs A B ->	0 0	0 1	1 1	1 0	Making Group those have same entry in each column
State a	b, -	<u>a</u> , 0	d, -	-, -	1 st Group (Because b, a, d & c present in all three state)
State b	<u>b</u> , 0	a, -	-, -	c, -	
State C	b, -	-, -	d, -	<u>c</u> , 0	
Sate d	-, -	a, 0	<u>d</u> , 1	e, -	2 nd Group (Because f, a, d & e present in all three state)
State e	f, -	-, -	d, -	<u>e</u> , -	
State f	<u>f</u> , 1	a, -	-, -	e, -	

Reduce Table:

Inputs A B ->	0 0	0 1	1 1	1 0	Remark/Step Follow
a, b, c	<u>b</u> , 0	<u>a</u> , 0	d, -	<u>c</u> , 0	Select stable state entry from each column in a group then select another state entry from each column in a same group.

d, e, f	f, 1	a, 0	d, 1	e, -	Select stable state entry from each column in a group then select another state entry from each column in a same group.
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Flow Table:

Consider $X1 \Rightarrow (a, b, c)$ and $X2 \Rightarrow (d, e, f)$

Inputs A B ->	0 0	0 1	1 1	1 0	Remark/Step Follow
X1	X1, 0	X1, 0	X2, -	X1, 0	Mark X1 & X2, according to a, b, c, d, e & f belongs with X1 & X2.
X2	X2, 1	X1, -	X2, 1	X2, 1	

Transition Table:

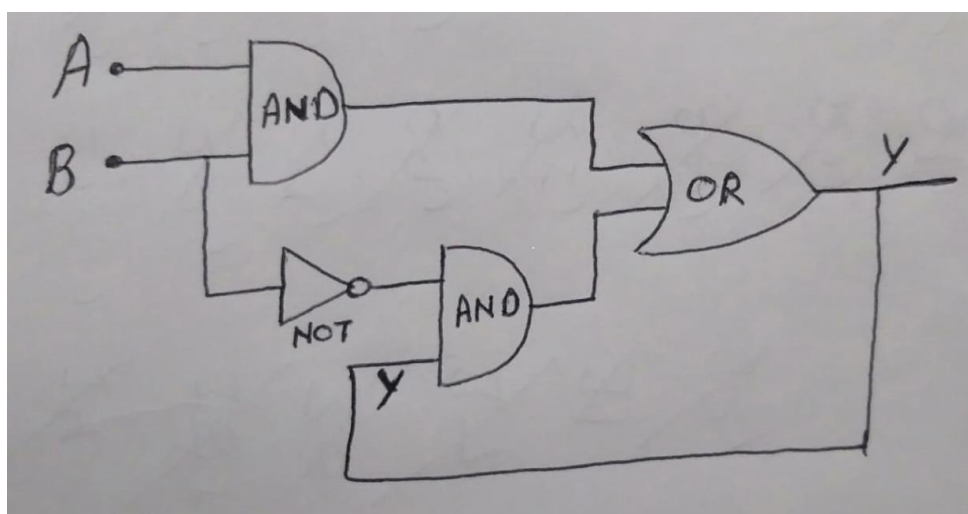
Consider $X1 \Rightarrow 0$ and $X2 \Rightarrow 1$

Inputs A B ->	0 0	0 1	1 1	1 0	Remark/Step Follow
Y = 0	0	0	1	0	Write X1 & X2 value, which are considered. Solve by K-Map and find function.
Y = 1	1	0	1	1	

1st function = $A.B$ and 2nd function = $Y.B'$

Function $Y = A.B + Y.B'$

Logic diagram:



Implement using SR Latch:

Start From Transition Table:

Inputs A B ->	0 0	0 1	1 1	1 0	Remark/Step Follow
Y = 0	0	0	1	0	Using SR Latch excitation table.
Y = 1	1	0	1	1	

Use SR Latch Excitation Table:

Present State	Next State	S	R
Q_n	Q_{n+1}		
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

For Find S Function:

Inputs A B ->	0 0	0 1	1 1	1 0	Function
Y = 0	0	0	1	0	Solve by K-Map and find function and Find Function for <u>S</u> = A.B
Y = 1	X	0	X	X	

For Find R Function:

Inputs A B ->	0 0	0 1	1 1	1 0	Function
Y = 0	X	X	0	X	Solve by K-Map and find function and Find Function for <u>R</u> = A'.B
Y = 1	0	1	0	0	

Logic Diagram: Using SR Latch with NOR Gate.

